Department of Computer Engineering

University of Peradeniya

CO543: Image Processing - Lab 7 (2025)

Transfer Learning & Fine-Tuning

Instead of learning from scratch, you will reuse a vision network trained on millions of images during this laboratory.

Task 1 — Use Pretrained CNN as Fixed Feature Extractor

Load a pretrained CNN (e.g., ResNet18). Replace the final layer with a new classifier for your dataset. Freeze all earlier layers and train only this classifier. Report training/validation accuracies and briefly describe how the frozen layers act as generic feature extractors.

```
for p in model.features.parameters():
    p.requires_grad = False
```

More to think:
If your dataset is simple and small, do you think this setup is enough?

Task 2 — Fine-Tune the Last Block

Unfreeze only the last convolutional block (keep earlier layers frozen) and train again with a smaller learning rate. Report the new performance and note any changes in convergence rate or overfitting signs.

```
for p in model.last_block.parameters():
    p.requires_grad = True
```

More to think:
Why adjust only higher layers instead of all layers at once?

Task 3 — Compare & Interpret

In your report, compare fixed-feature vs partial fine-tuning: accuracy, stability, and training behavior. Explain when freezing works well (similar domains, small data) and when fine-tuning is necessary (different textures/shapes, domain shift).

More to think:
Why can fine-tuning on a very small dataset cause overfitting faster than using the frozen model?

Final Reflection

Describe in one paragraph how transfer learning speeds up development and where it might fail.

Submission

Submit **one Jupyter notebook (.ipynb)** with:

- *Code for each task (concise, no redundant cells)*
- *Output images shown in-notebook*
- *Short answers where required*
- *Final reflection at the end*
- ***E_21_xxx_Lab_yy.ipynb** (xxx = reg no., yy = lab number)*
